```r
library(tidyverse) |> suppressPackageStartupMessages()
```

```
Warning: package 'tidyverse' was built under R version 4.5.2

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2
```

```r
library(plotly) |> suppressPackageStartupMessages()
```

```
Warning: package 'plotly' was built under R version 4.5.2
```

```r
library(colorspace) |> suppressPackageStartupMessages()
dark_steelblue <- darken("steelblue4", 0.2)
sat_blue <- '#009bdb'

theme_eda <- theme(
  # Overall text
  text = element_text(family = "Helvetica", color = "black"),

  # Plot background
  plot.background = element_rect(fill = "white", color = NA),
  panel.background = element_rect(fill = "grey98", color = NA),

  # Gridlines
  panel.grid.major = element_line(color = "grey85"),
  panel.grid.minor = element_line(color = "grey90"),

  # Titles
  plot.title = element_text(size = 20, face = "bold", color = dark_steelblue, hjust = 0.5, ma
  plot.subtitle = element_text(size = 14, color = "grey30", hjust = 0.5, margin = margin(b =

  # Axis
```

```r
  axis.title = element_text(size = 14, face = "bold", color = dark_steelblue),
  axis.text = element_text(size = 12, color = "grey20"),
  axis.ticks = element_line(color = "grey70"),

  # Legend
  legend.background = element_rect(fill = "grey98", color = NA),
  legend.key = element_rect(fill = "grey98"),
  legend.title = element_text(size = 14, face = "bold", color = dark_steelblue),
  legend.text = element_text(size = 12, color = "grey20"),
  legend.position = "top",
  legend.key.size = unit(1.2, "lines"),

  # Facets
  strip.background = element_rect(fill = "steelblue3", color = NA),
  strip.text = element_text(size = 14, face = "bold", color = "white"),

  # Margins
  plot.margin = margin(10, 10, 10, 10)
)

# Apply globally
theme_set(theme_eda)
```

```r
admin_df <- read_csv("cleaned_admin_df.csv")
```

```
Rows: 287 Columns: 15
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (4): INSTNM, CITY, STABBR, ZIP
dbl (11): UNITID, HIGHDEG, ST_FIPS, LATITUDE, LONGITUDE, LOCALE, ADM_RATE, S...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
admin_test_df <- read_csv("admin_test_full.csv")
```

```
Rows: 287 Columns: 18
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (6): INSTNM, CITY, STABBR, ZIP, Policy_2026, Is_policy_temporary
dbl (12): UNITID, HIGHDEG, ST_FIPS, LATITUDE, LONGITUDE, LOCALE, ADM_RATE, S...
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#testing plotly charts

#fitting a linear model

fig <- plot_ly(admin_df,
               x = ~SAT_AVG, y = ~C150_4,
               type = 'scatter', mode = 'markers',
               marker = list(color = sat_blue),
               name = 'Schools') |>
  layout(
    title = list(
      text = "Relationship Between SAT Average Scores and 6-Year Completion Rates",
      font = list(size = 18, color = "black", family = "Arial"),
      x = 0.5  # centers the title like hjust = 0.5
    ),
    xaxis = list(
      title = list(text = "SAT Average Score for Accepted Students", font = list(size = 14, c
      tickfont = list(size = 12, color = "black", family = "Arial")
    ),
    yaxis = list(
      title = list(text = "Proportion of Undergrad Students Who Graduate in 6 Years", font =
      tickfont = list(size = 12, color = "black", family = "Arial"),
      tickformat = ".0%"
    ),
    legend = list(
      title = list(text = "", font = list(size = 14, color = "black")),
      font = list(size = 12, color = "black")
    )
  )
```